



INTE130 – Object Oriented Programming

Designing a One Person Company Using a Multi – Agent System.

Submitted : Part A and Part B

Prepared by :

Sultan Khalid AlYarabi – 2318592

Malik Marzzoq Al Rahbi – 2527871

Khamis Hamed AlOwaisi – 2422670

Mohamed Naeem Al Gheilani – 2526168

Instructor Name:

Dr. Akbar Khan

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Part A – Understanding One Person Company Systems

Frist : One Person Company Concept:

A One person company is a business that is owned and managed by one person. The owner is responsible for running the company and making decisions. This type of company helps people start their own business without needing many employees.

The company can use digital systems to help in daily work. These systems make many tasks easier and faster.

Role of Digital Systems in the Business

Digital Systems help the owner in different ways such as :

Managing business tasks more easily

Saving time and effort

Organizing company data

Helping the need for many workers

Improving work efficiency

Second : Architecture of one person company:

A one person company usually includes the following:

Founder: the founder is the main person in the company . He manages the business, makes decisions, and controls the company operations.

Digital Agents : Digital Agents are programs or systems that help the owner complete tasks automatically. They support the daily work of the company.

Business Tasks:

The company has different tasks such as:

Financial management

Customer service

Marketing

Data management.

Monitoring operations.

System components:

The System contains different tools and platforms

That help the company work properly . These may

Include:

Registration systems

Communications tools

Business management platforms
Financial systems

Third: Role of Multi Agent Systems:

A multi Agent system helps divide the work between different agents. Each agent performs a specific task. This helps the owner complete many tasks at the same time.

Using multi agent systems can :

Reduce workload

Save time

Improve work speed

Increase efficiency

Help one person manage the whole company more easily.

Fourth: Programming perspective:

In OOP , The one person company system can be represented using different classes.

Class Founder: This class represents the company owner who makes decisions and manages the company.

Class Agents: This class represents the digital agents . Each agent has its own job and function.

Class Company: This class represents the company itself and stores the company information and operations.

Class Tasks : This class represents the different tasks that are performed inside the company.

Part B – Business Idea and Project Proppsal:

Business Name:

CyberShield System

Business Background:

CyberShield System is one person company that provides cybersecurity services for a small and medium business or company. The puropse of the system is to help companies protect there system and networks and data from cyber threts.

The system can monitor acivites , detect security weaknesses and provide security suggestions . Many small companies can not afford a full cybersecurity team , so this system provides a simple and usful solution using digital agents managed by founder.

3- Target Customer:

The target customer for CyberShield are:

Small and medium sized businesses

Startups

E commere companies

These businesses need cybersecurity protection but some of them do not have enough budget to hire a full security team. Because of this, they can use CyberShield as a simpler and more affordable solution.

4-Why the Business Can be managed by one person?

This business can be managed by one person because most tasks are handled by digital agents. The founder supervises the system and makes the final decision.

The agents can help with:

Monitoring threats

Checking vulnerabilities

Preparing reports

Suggesting solutions

This reduces the amount of manual work and makes the system easier to manage.

5-Proposed agents:

Financial agents: this agent helps manage costs , calculate service prices and prepare simple financial reports.

Threat monitoring agent:

This agent monitors the system and detects suspicious activities such as unusual login attempts or unauthorized access.

Vulnerability analysis agent:

This agent scans the system for weaknesses like weak passwords or outdated software and gives suggestions for improvement.

Incident Response Agent

This agent suggests actions that should be taken when a security issue happens , such as blocking access or changing passwords.

Reporting Agent:

This agent prepares reports about the system status , security issues and recommendations.

6- Example interaction:

Frist , the founder requestess a system security cheak .
than the monoitoring agent checks the system and
detecta any suspicious activity.

After that the vulnerability anlysis scans the system for
weaknesses . the incident response agent than suggests
possible solutions for the detected problem.

Finally , the reporting agent perapers a report , and the
founder makes the final decision based on the result.